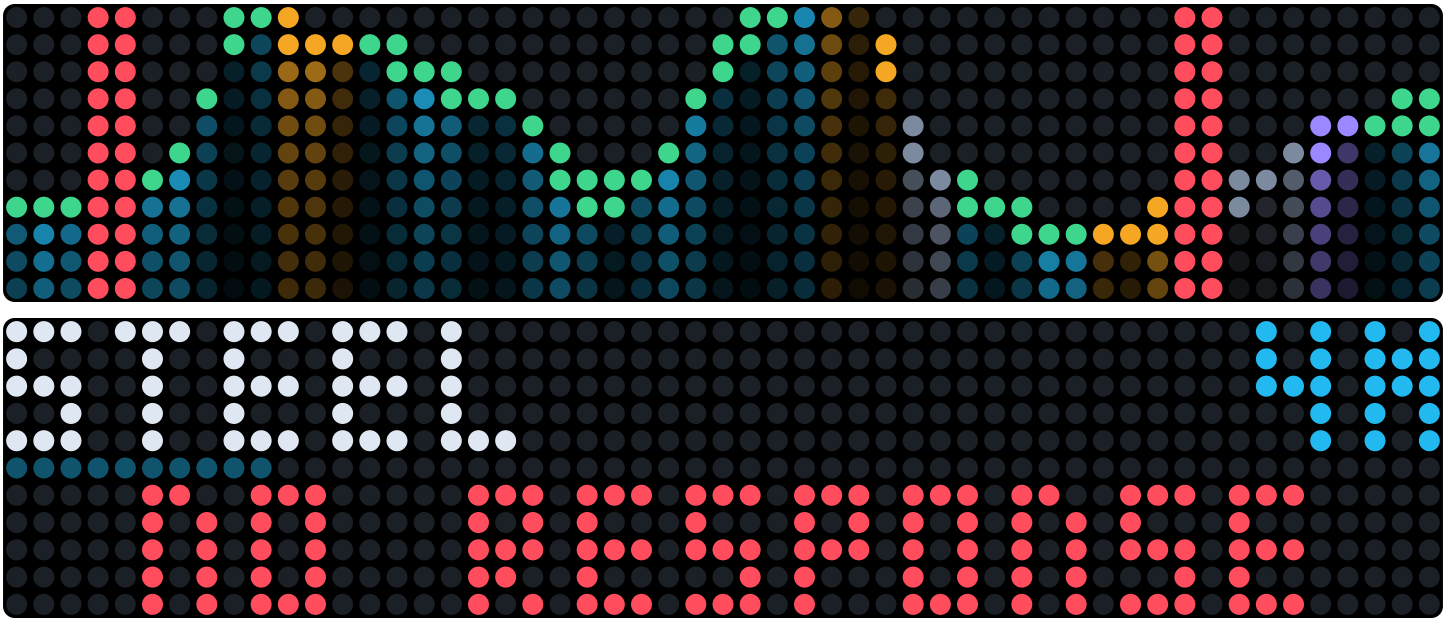
Panel Screen Guide

Every screen on the wall panel: what it is, what each part shows, how the screens are grouped, and how to set them up.



FOR
Anyone who looks at the panel and wonders what it is saying.

SYSTEM
Galactic Unicorn panel on lithium
Screens v2, November 2026

EDITION
1.0 · 5 November 2026
US Letter · screen and print

CONTENTS

What is in this guide

1 **How the panel is organised**
Four groups, sixteen screens, one rotation.

2 **Fleet: what is up?**
Host Tiles, Roster, Pools, Head Count.

3 **Traffic: what is moving?**
Swim Lanes, WAN and LAN, Top Talkers, Flow Map.

4 **Trouble: what is wrong?**
Alerts, Fault Board, Reachability, Alert History.

5 **Resources: how hard are they working?**
Host Detail, CPU and Memory, Trends, Aurora.

6 **Screens that interrupt**
Alarm, screen name, help, and lost data.

7 **Every setting**
Panel buttons and the dashboard Panel page.

START HERE

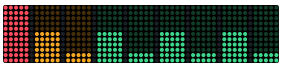
How the panel is organised

Sixteen screens, in four groups of four. Each group answers one question. The panel tours every screen that is switched on, and the A, B, C and D buttons jump to a group.

A

Fleet

What is up?



A0 Host Tiles



A1 Roster



A2 Pools

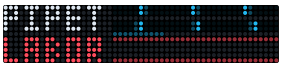


A3 Head Count

B

Traffic

What is moving?



B0 Swim Lanes



B1 WAN and LAN



B2 Top Talkers



B3 Flow Map

C

Trouble

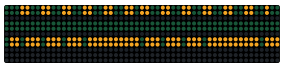
What is wrong?



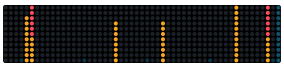
C0 Alerts



C1 Fault Board



C2 Reachability



C3 Alert History

D

Resources

How hard are they working?



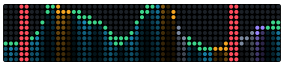
D0 Host Detail



D1 CPU and Memory



D2 Trends



D3 Aurora

Every picture in this guide is a real frame from the panel's drawing code, using test data. Your host names and numbers will differ.

Reading any screen

- **Colour carries the meaning.** The same colour means the same thing on every screen.
- **Text never overlaps.** A screen holds at most two rows of 13 characters. Longer lists page every 3 s, and a faint bar between the rows shows the progress. Host names are short names, never full domain names.
- **Everything that moves is measured.** Nothing is animated for effect. If a thing is not measured, the panel shows it as no data rather than as zero.

COLOUR	MEANS
White	Names and the main fact on a row
Blue	Second fact: rates, ages, CPU. Also the page bar
Violet	Memory, upload in a sparkline, a Wi-Fi link, or a host in maintenance
Green	Healthy. On traffic screens, a host's own network link
Amber	Degraded, a warning, or a reading well above its usual level
Red	Down, lost, or critical. Never used for decoration
Grey	Unknown: no measurement, which is not the same as zero

GROUP A

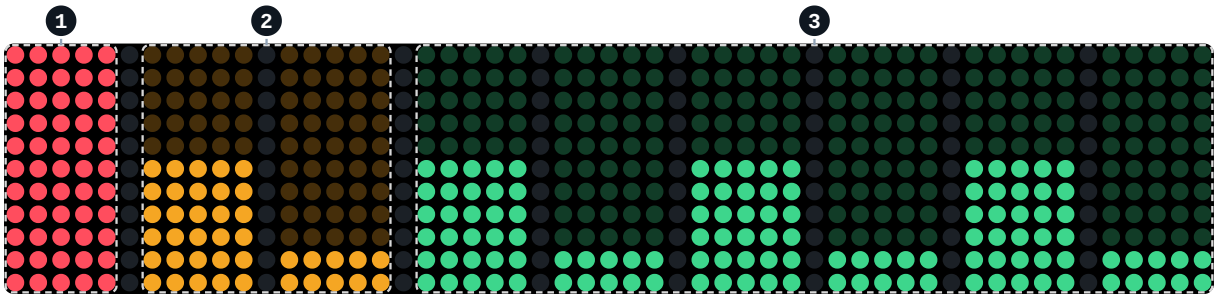
Fleet: what is up?

How many hosts are up and which is the worst. Every screen here counts the same hosts.

A0 Host Tiles

TILE IS HOST / FILL IS LOAD

The whole fleet at a glance. Every host owns a tile, and the tiles always fill the panel.



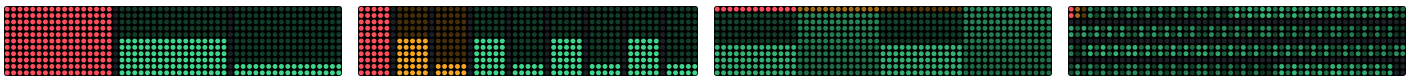
1. **Worst host, on the left.** Tiles run worst first, so trouble reads from the left. Solid red is down.
2. **Degraded hosts.** Amber. The lit part of a tile is its CPU load; the dim part is what is free.
3. **Healthy hosts.** Green, again filled to their load.

CONFIGURE

Rotation on or off, dwell weight ¼× to 10×. Default on, 1×. The tile size is automatic.

WITH NOTHING TO SHOW

Shows **NO HOSTS**.



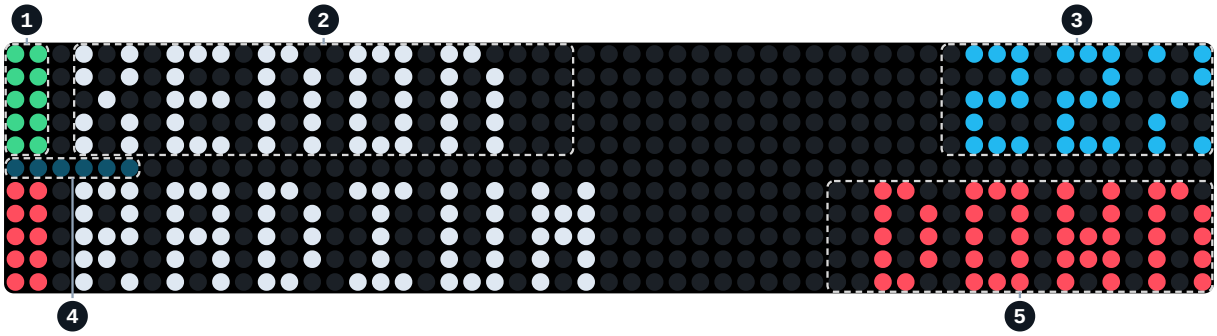
3 hosts | 9 hosts | 40 hosts | 191 hosts

The same screen at four fleet sizes. Three hosts are big blocks and 191 a fine mosaic, so a small fleet is never a dull screen with a few dots.

A1 Roster

HOST LIST / WORST FIRST

Hosts by name, worst first, two at a time.



1. **State block.** The host's colour, the same as its tile.
2. **Name.** The short host name, never the full domain name.
3. **Load.** CPU percent. Blue when the host is healthy, and the host's own colour when it is not.
4. **Page bar.** A faint bar in the gap between rows moves left to right through the pages.
5. **State word.** Replaces the load when the load means nothing: `DOWN`, `MNT`, or `?` for unknown.

CONFIGURE

Rotation on or off, dwell weight ¼× to 10×. Default on, 1×. Paged, so the panel keeps it up long enough to show each page, 3 s a page, up to 19 s.

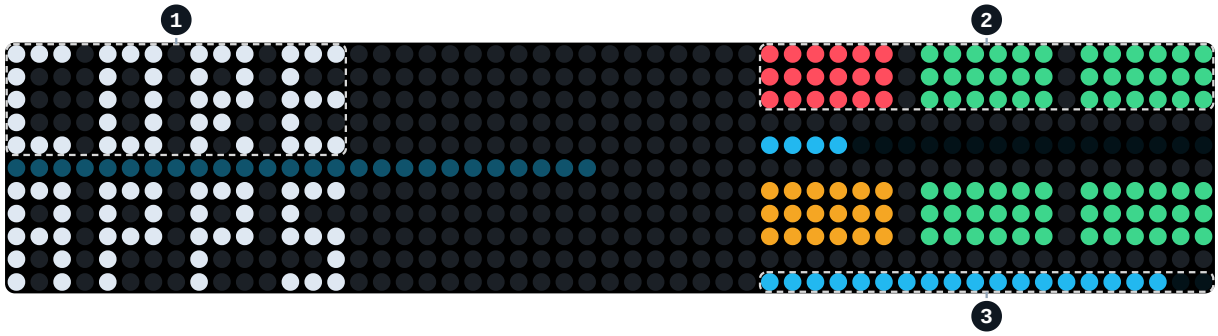
WITH NOTHING TO SHOW

Shows **NO HOSTS**.

A2 Pools

EACH POOL / HOSTS LOAD

Groups of hosts by the job they do, two at a time.



1. **Pool name.**
2. **Hosts in the pool.** One block per host in its state colour. Past 20 hosts the blocks become proportional runs.
3. **Pool load.** The thin line beneath grows with the pool's average CPU.

CONFIGURE

Rotation on or off, dwell weight ¼× to 10×. Default on, 1×. Pool membership comes from each asset's pool in the dashboard. Paged, so the panel keeps it up long enough to show each page, 3 s a page, up to 19 s.

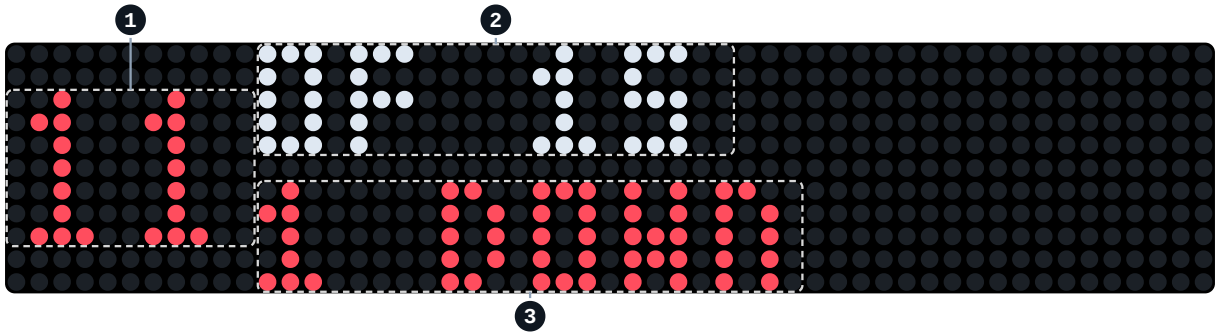
WITH NOTHING TO SHOW

An empty pool shows its name alone.

A3 Head Count

HOSTS UP / AND WORST NOW

The one number you can read from across the room.



1. **Hosts up.** Big digits in the colour of the worst state: green when all is well, red when any host is down.
2. **Out of how many.**
3. **The worst thing right now,** in the same colour: `1 DOWN`, `2 DEGRADED`, or `ALL OK`.

CONFIGURE

Rotation on or off, dwell weight ¼× to 10×. Default on, 1×.

WITH NOTHING TO SHOW

Shows **NO HOSTS**.

GROUP B

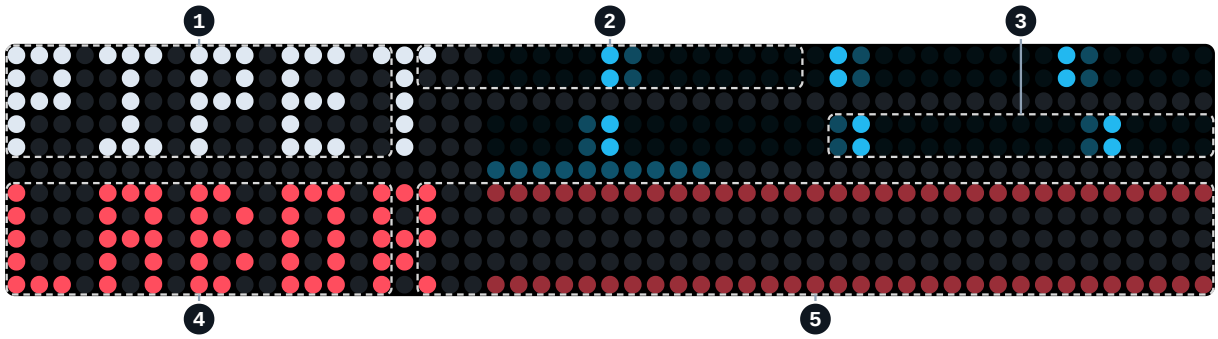
Traffic: what is moving?

Measured rates only. Rates share one log scale, so a busy internet line and a quiet local link are both visible.

B0 Swim Lanes

LINK TRAFFIC / DOTS ARE RATE

Traffic on each network link, two at a time. Every link keeps its own lane and all share a log scale, so the internet cannot drown out a quiet LAN link.



1. **Link name.** The internet is `WAN`, and a failover line `WAN2`.
2. **Upper stream, away from the internet.** More traffic means denser, faster dots. The colour is the kind of link: white internet, blue switch to switch, violet Wi-Fi access point, green host.
3. **Lower stream, toward the internet.** Sparse grey dots that hold still mean no measurement, which is not the same as zero traffic.
4. **A down link's name** turns red.
5. **Its lane** becomes two red rails.

CONFIGURE

Rotation on or off, dwell weight ¼× to 10×. Default on, 1×. Links come from the UniFi poller and host agents, up to 12. Paged, so the panel keeps it up long enough to show each page, 3 s a page, up to 19 s.

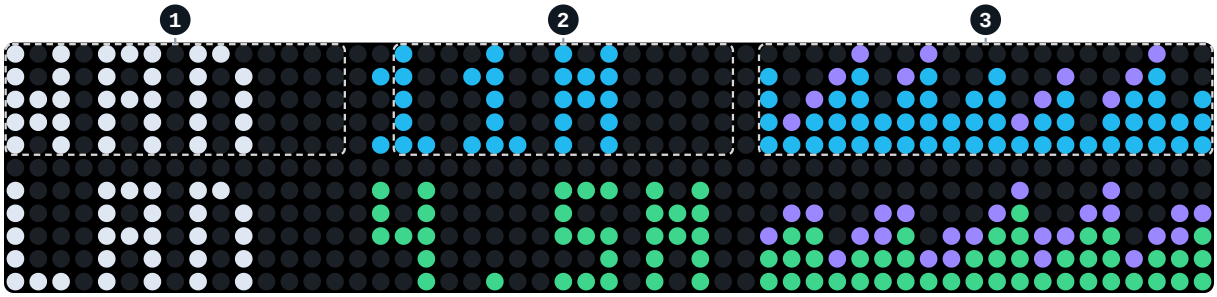
WITH NOTHING TO SHOW

Shows **NO LINK DATA**.

B1 WAN and LAN

INTERNET AND / LOCAL TRAFFIC

The internet and the local network, side by side, each on its own scale.



1. **Which network.** `WAN` is the internet; `LAN` below it is everything local.
2. **Total rate right now.** Download plus upload.
3. **Recent history.** One column per sample, newest on the right. Blue or green is download, violet stacked above it is upload. The strip scales to its own peak, so read its shape rather than its height.

CONFIGURE

Rotation on or off, dwell weight ¼× to 10×. Default on, 1×. Each history is kept on the panel and is empty after a restart.

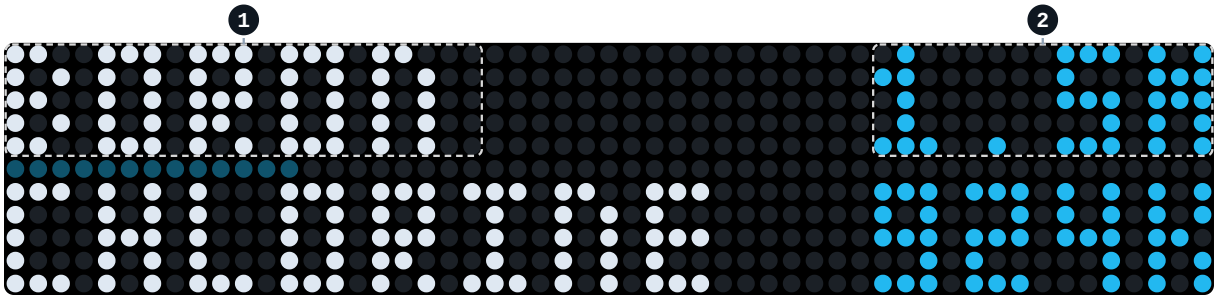
WITH NOTHING TO SHOW

Shows **NO RATE DATA**.

B2 Top Talkers

BUSIEST HOSTS / BY TRAFFIC

The busiest hosts by measured traffic, two at a time.



1. **Host.** Busiest first.
2. **Rate.** Download plus upload on that host's own network card.

CONFIGURE

Rotation on or off, dwell weight ¼× to 10×. Default on, 1×. Paged, so the panel keeps it up long enough to show each page, 3 s a page, up to 19 s.

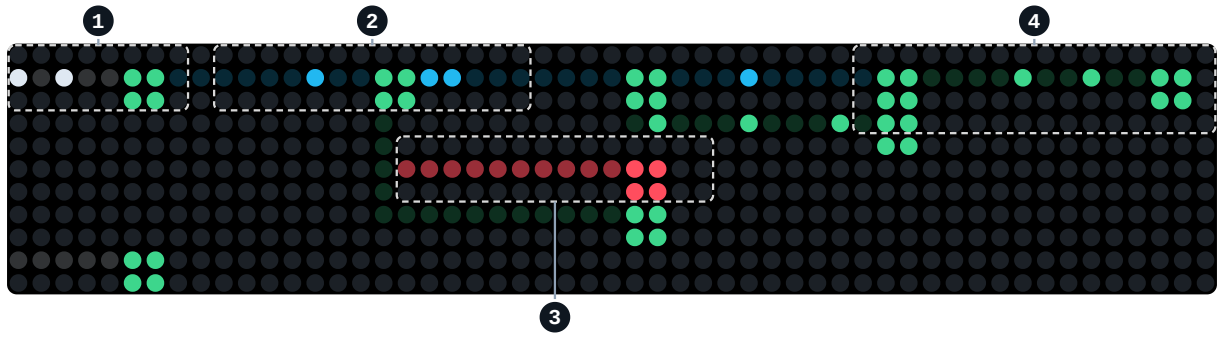
WITH NOTHING TO SHOW

Shows **NO HOST RATES**.

B3 Flow Map

NETWORK TREE / DOTS ARE RATE

The network as it is cabled: the internet on the left, and each device at its real depth.



1. **The internet** comes in from the left edge.
2. **A device** is a small block in its state colour. Dots move along its uplink at that link's measured rate, using the same link colours as Swim Lanes.
3. **A down link** is drawn in solid red, with no dots.
4. **Deeper devices** sit further right. A link with no traffic, or no measurement, has no dots.

CONFIGURE

Rotation on or off, dwell weight ¼× to 10×. Default on, 1×. Depth comes from the uplinks UniFi reports, so it changes when a device moves.

WITH NOTHING TO SHOW

Shows **NO LINK DATA**.

GROUP C

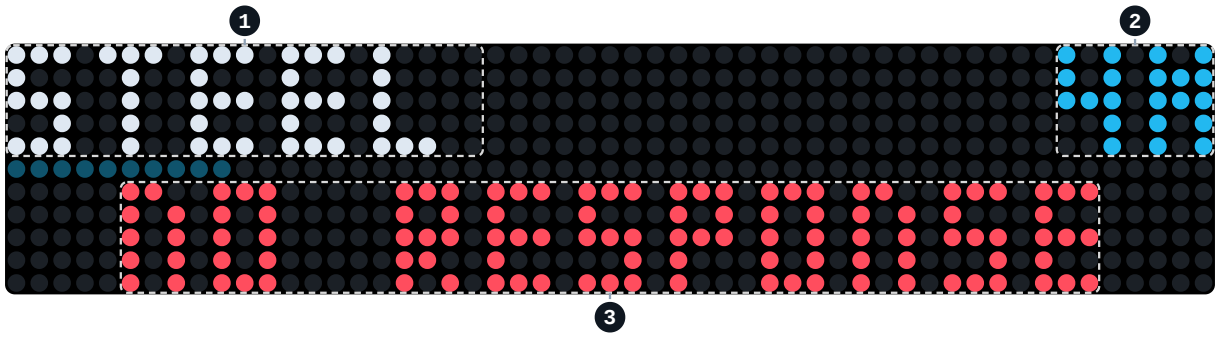
Trouble: what is wrong?

Open problems, by host and by kind, and how they have changed. One problem counts once, however many monitors see it.

C0 Alerts

OPEN ALERTS / WORST FIRST

One open problem at a time, in plain words. Up to eight show in turn, then a count of the rest.



1. **Host.**
2. **How long it has been open.** In minutes, hours or days.
3. **What is wrong,** in the severity's colour. For example `NO RESPONSE`, `HTTPS DOWN`, `DISK READONLY`, `2 SVCS FAILED`. The last page is `+10 MORE` over `SEE DASHBOARD`.

CONFIGURE

Rotation on or off, dwell weight ¼× to 10×. Default on, 1×. Paged, so the panel keeps it up long enough to show each page, 3 s a page, up to 19 s.

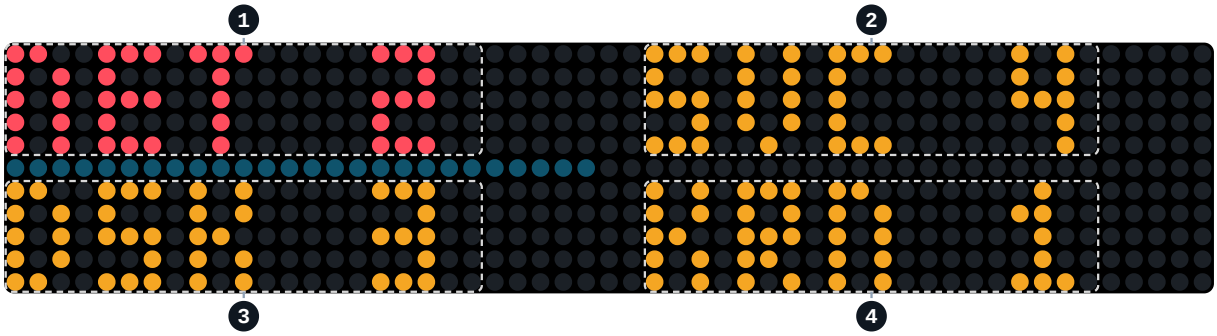
WITH NOTHING TO SHOW

Shows **ALL CLEAR** in green.

C1 Fault Board

OPEN FAULTS / BY KIND

Open problems counted by kind, so you can see at once what sort of day it is. Green zeros are information too.



1. **NET.** Something is not answering.
2. **SVC.** A service has failed on a host.
3. **DSK.** A disk is read-only, failing or missing.
4. **KRN.** The kernel is logging errors.

CONFIGURE

Rotation on or off, dwell weight ¼× to 10×. Default on, 1×. Anything else shows on a second page as `OTHER`.

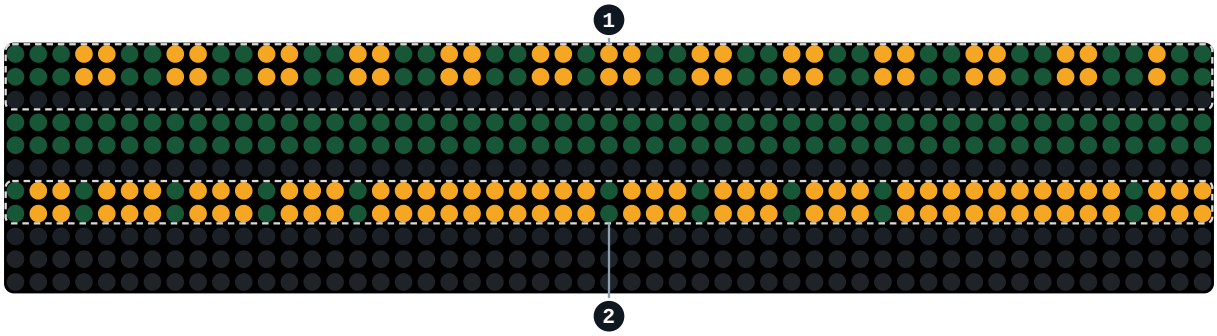
WITH NOTHING TO SHOW

Green zeros.

C2 Reachability

PING HISTORY / RED IS LOSS

One row per probed host, newest on the right. Rows share the height, so three hosts make thick bands.



1. **A healthy row.** Dim green is at the host's own usual response time. Amber is at least twice that. Red is a lost probe.
2. **Spikes stand out against each row's own baseline,** so a host that is always slow is not always amber. A host that has never answered stays grey.

CONFIGURE

Rotation on or off, dwell weight ¼× to 10×. Default on, 1×. Each history is kept on the panel and is empty after a restart. Targets come from the monitors, up to 11.

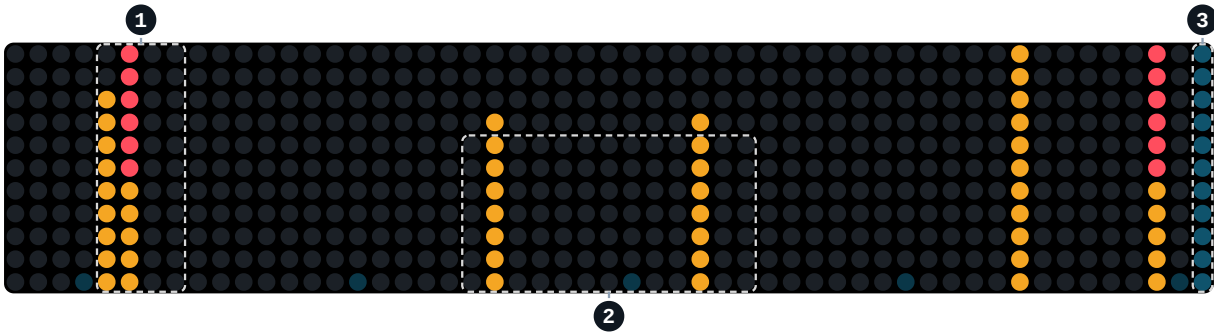
WITH NOTHING TO SHOW

Shows **NO PROBE DATA**.

C3 Alert History

ALERTS 26 HRS / NOW AT RIGHT

How many alerts fired in each half hour, across the last 26 hours.



1. **A bad half hour.** Height is on a log scale, so one bad night does not flatten the rest. Red, critical, sits on top of amber, warning.
2. **Single firings.** Dim ticks along the floor mark every six hours.
3. **Now.** The blue line at the right edge.

CONFIGURE

Rotation on or off, dwell weight ¼× to 10×. Default on, 1×.

WITH NOTHING TO SHOW

Shows **NO HISTORY**.

GROUP D

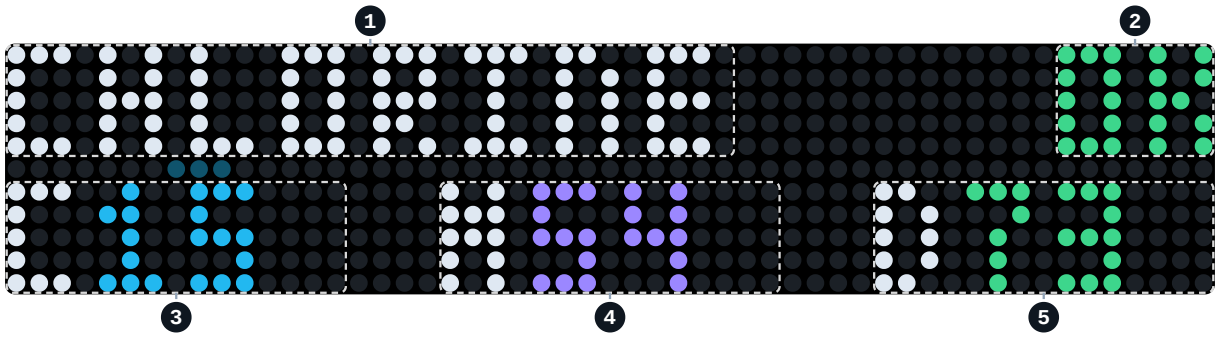
Resources: how hard are they working?

CPU, memory and disk per host, recent trends, and one ambient screen that still tells you the truth.

D0 Host Detail

ONE HOST / CPU MEM DISK

One host at a time, worst first, with its three readings.



1. **Name.**
2. **State word.** `OK`, `DEG`, `DOWN`, `MNT`.
3. **C: CPU percent.**
4. **M: memory percent.**
5. **D: disk percent.** Each number turns amber at 75% and red at 90%.

CONFIGURE

Rotation on or off, dwell weight ¼× to 10×. Default on, 1×. Paged, so the panel keeps it up long enough to show each page, 3 s a page, up to 19 s.

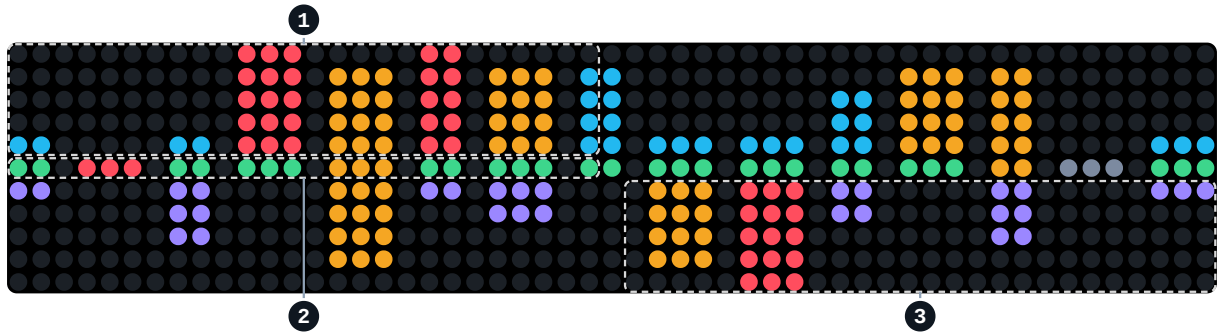
WITH NOTHING TO SHOW

A host reporting nothing shows **NO READINGS**.

D1 CPU and Memory

UP IS CPU / DOWN IS MEM

A butterfly chart: every host's CPU and memory at once.



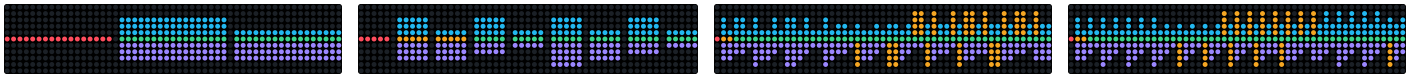
1. **CPU rises from the centre.** Blue, turning amber at 75% and red at 90%.
2. **The centre row is each host's state.** A host that is down has no readings, only this row.
3. **Memory falls from the centre.** Violet, with the same warning colours.

CONFIGURE

Rotation on or off, dwell weight ¼× to 10×. Default on, 1×. Column width is automatic.

WITH NOTHING TO SHOW

Shows **NO HOSTS**.



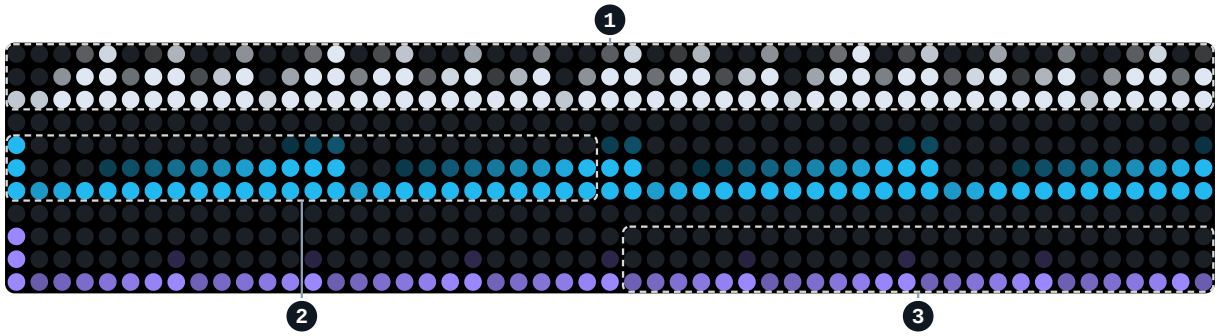
3 hosts | 9 hosts | 40 hosts | 191 hosts

Columns take whatever width the fleet leaves.

D2 Trends

TRAFFIC CPU / AND PING

The last 106 seconds, as three strips. Each scales to its own window, with a floor, so a calm network has visible shape without looking like a crisis.



1. **Internet traffic.**
2. **Mean CPU across the fleet.** Floor of 40%.
3. **Ping.** Floor of 10 ms.

CONFIGURE

Rotation on or off, dwell weight ¼× to 10×. Default on, 1×. History is kept on the panel and is empty after a restart.

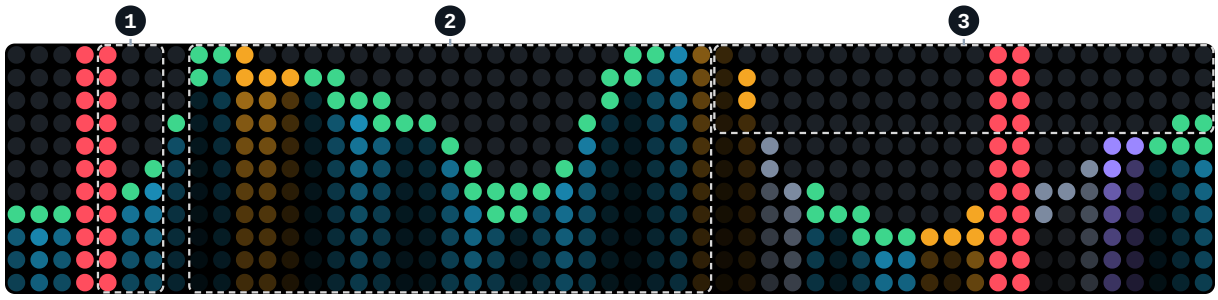
WITH NOTHING TO SHOW

Strips stay empty until samples arrive.

D3 Aurora

AMBIENT FLEET / RED IS FAULT

The ambient screen, and it is still a real reading. It reads as a picture from across the room and shows a fault at once.



1. **A failed host is a red pillar.**
2. **Each host owns a band of columns.** The curtain's height is its load, and its colour is its state.
3. **Drift speed is real traffic.** With no traffic, the curtain stands still. The rays beneath are texture and carry no reading.

CONFIGURE

Rotation on or off, dwell weight ¼× to 10×. Default on, 1×.

WITH NOTHING TO SHOW

Shows **NO HOSTS**.

EVERYWHERE

Screens that interrupt

These five can appear over or instead of any screen.



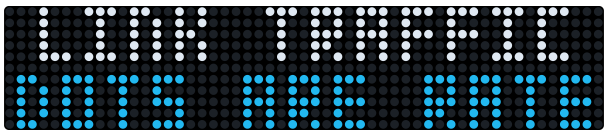
Alarm

A critical condition is live. The host and the fault cover the screen, red rails run down both edges, and a two-tone sounds. Any button acknowledges it. The fault stays open until it is fixed.



Screen name

Shown for one second as a new screen comes up, over the dimmed screen. It tells you where you are without taking space from the screen.



Help

Press `Vol+` for two lines on the screen showing. It closes on a second press, on any other button, or after 30 s. It does not open during an alarm.



No data feed

The panel has stopped receiving. It says so plainly because whatever it was showing is now old. Check `solari-panel.service` on lithium, then the USB cable.



No hosts

The panel is connected, but the feed holds no hosts yet. Normal for a moment after a restart.

REFERENCE

Every setting

Everything is set from the panel's own buttons or from the **Panel** page of the dashboard. Operators and admins can change settings; viewers can look.

On the panel

BUTTON	WHAT IT DOES
`A` `B` `C` `D`	Jump to that group. Press the same button again for its next screen.
`Lux+` `Lux−`	Brightness, 5% a step. This turns off automatic brightness until it is turned back on from the dashboard.
`Vol+`	Help for the screen showing.
`Vol−`	Alarm tone quieter, 10% a step. There is no louder button; a power cycle restores it.
`Zzz`	Sleep the display. Any button wakes it, and so does a new alarm.
Any button during an alarm	Acknowledges the alarm, and does nothing else.

On the dashboard

SETTING	WHAT IT DOES	DEFAULT
Theme, Screen	Jump to a group or a screen, as the buttons do.	Stays in the tour from there
Brightness, AUTO	Set a percentage, or let the light sensor drive it.	AUTO
Dwell (base)	How long each screen stays up: 3, 6 or 30 s.	6 s
Rotation, per screen	On or off. Off skips it in the tour and on the group buttons.	On
Dwell weight, per screen	Multiplies the base dwell: ¼×, ½×, 1×, 2×, 5×, 10×.	1×
Acknowledge	Silences the current alarm, as a button does.	–
Wake, Sleep	Same as `Zzz`, from anywhere.	Awake

How the settings behave

- **They survive a restart.** The per-screen settings are saved to the panel itself about two seconds after a change. The dashboard shows a setting as in force only once the panel confirms it.

- **A paged screen can stay longer than its dwell.** It gets enough time to show its pages, up to 19 s.
- **You cannot blank a group.** If every screen in a group is off, the panel treats them all as on rather than show a black wall. To blank the panel, use **Sleep**.

> **GOING DEEPER**
> The *Status Panel Manual* in the project's `docs/panel` folder covers hardware and the alarm path in depth, and the *Quick Start Guide* on this portal shows how to answer an alarm step by step.